

The possibility to observe QCD instanton

I.I. Balitsky¹ and M.G. Ryskin

Petersburg Nuclear Physics Institute, 188 350 Gatchina, St. Petersburg, Russian Federation

Received 29 June 1992; revised manuscript received 9 September 1992

If, at high energies, the QCD-instanton-induced cross section reaches its unitarity limit, this process should be considered as the main source of the large transverse energy events. The events of such a type should be observed already at Sp̄pS or FNAL colliders in the range of $E_T > 100$ –200 GeV. However, if the emission of a large number of gluons compensates no more than one half of the exponential suppression of the instanton amplitude, it will be not so easy (but at $E_T \sim 30$ GeV still hopeful) to select the events induced by the QCD instantons.

1. The recent interest in the problem of baryon number violation at high energies [1] has focused attention on nonperturbative approaches of the anomalous scattering. A nonperturbative configuration of field instanton [2] induces processes with a rather large cross section. This is connected with the fact that exponential suppression of the instanton amplitude, $\exp(-2\pi/\alpha)$, is compensated by the growth with energy of the high multiplicity cross sections, $\sigma_n \sim (E^{4/3}/g^{2/3})^n$. Indeed, from the point of view of Feynman graphs the instanton reveals itself as a system of point-like vertices ($\tilde{\mathcal{L}}_n \phi^n$) which describe the emission of a large number (n) of vector- and Higgs bosons (provided the latter is in our theory) with the amplitudes $\tilde{\mathcal{L}}_n \sim (E^2/\alpha)^{n/2}$. The cross section of the production of these multiparticle clusters increases exponentially with energy and available phase space. In the framework of QCD it looks natural that the nonperturbative instanton configuration of fields leads to the same production of multiparticle clusters at high energies. Of course, such a process does not look as an anomalous one. Instead of baryon number violation the QCD instanton violates the chirality only, but the S-wave isotropic point-like production of the large number of hard-gluon jets may be used as

a rather good trigger to select the instanton-induced events experimentally.

2. Let us estimate the scale of the cross section. As compared to the electroweak theory where the size of the instanton is determined by the mean value of the Higgs field ($\rho \sim 1/\nu$), the problem arising in QCD is the following. The radius of QCD instanton may be very large, up to $1/\Lambda_{\text{QCD}}$, but one does not know the behaviour of the cross section for the asymptotically large energy, $Ep \rightarrow \infty$. Therefore, we consider here two extreme scenarios:

(A) the cross section reaches the unitarity limit [3], $\sigma_0 = 16\pi/E^2$;

(B) the cross section reaches the maximal, $\sigma_{\text{max}} \sim \sigma_0 \exp(-2\pi/\alpha_s)$, and then decreases [4] with the further increase of Ep .

(A) Let us use the most popular assumption that at high energies the amplitude of the instanton-induced processes may be calculated within the one-instanton approximation using the unitarity and the action $S_{\text{int}} = S(\text{I}-\bar{\text{I}} \text{ interaction})$, calculated by the valley method [5], to sum up all the multigluons channels in the total cross section.

It is convenient to introduce the following notations: ρ_1 is the radius of the instanton I, ρ_2 is that of the anti-instanton $\bar{\text{I}}$, R is the distance between instanton and anti-instanton; the conformal coordinate $z = (R^2 + \rho_1^2 + \rho_2^2)/\rho_1\rho_2$.

Then the action S_{int} that describes the instanton-

¹ Present address: Physics Department of Pennsylvania State University, 104 Davey Laboratory, University Park, PA 16802, USA.

anti-instanton interaction may be written in the dipole-dipole approximation ^{#1}

$$S_{\text{int}} = \frac{4\pi}{\alpha_s} \frac{6}{z^2} + n_F \ln \frac{2^4}{z^3}, \quad (1)$$

where $n_F=3$ is the number of the light-quark types and the term $n_F \ln(2^4/z^3)$ takes into account the emission of light quarks due to the zero fermion mode in the instanton field.

The one-instanton action is equal to $S_0 = -2\pi/\alpha_s$ and, taking into account the determinant, one gets the factor

$$J_1 = 1.7 \times 10^{-3} \frac{d\rho}{\rho^5} \left(\frac{2\pi}{\alpha_s} \right)^6 \exp\left(-\frac{2\pi}{\alpha_s}\right). \quad (2)$$

The wave function of the initial gluon is given by the Fourier transform of the instanton field,

$$A_\mu^a(q^2) = \int d^4x \exp(iqx) \left(-\frac{2i}{g} \frac{\eta_{\mu\nu}^a x_\nu \rho^2}{x^2(x^2 + \rho^2)} \right) \\ = \sqrt{\frac{4\pi}{\alpha_s}} \pi \frac{\rho^2}{q^2} \eta_{\mu\nu}^a q_\nu. \quad (3)$$

(Here we have used the singular gauge and put $qp \ll 1$; the Fourier transform calculated within the valley that reproduces the action $S_{\text{int}} = (4\pi/\alpha_s)(6/z^2)$ differs from eq. (3) by small corrections of the order of $\sim \rho^2/R^2$.)

Another factor, $\int \exp(iER) d^4R$, is connected with the fact that the initial energy E but not the instanton-anti-instanton distance R is fixed in the experiment.

If the QCD coupling, α_s , is small enough ($4\pi/\alpha_s \gg 1$) one can calculate all the integrals using the method of stationary point. The saddle points are

$$R=0, \quad ER_0 = 24 \frac{4\pi}{z_0^2} \frac{z_0 - 2}{z_0^3}, \quad \lambda = \rho_1 - \rho_2 = 0, \quad (4)$$

and the relative instanton-anti-instanton orientation corresponds to the dipole-dipole interaction (the essential values of parameter $16(4\pi/\alpha_s z^2) \sim 8-12$ is rather large to use the saddle point method averaging over the orientation of instantons).

^{#1} One can use the action $S_{\text{int}}(z)$ calculated with more accuracy by the valley method (see refs. [3,6]) but the expression (1) is enough for our purpose as the essential values of $z \geq 5$ are rather large.

Finally, the cross section takes the form

$$\sigma_1 = \sigma_0 \left[\frac{1.7 \times 10^{-3}}{16\pi} \exp\left(-\frac{2\pi}{\alpha_s}\right) \left(\frac{2\pi}{\alpha_s}\right)^8 \frac{\pi^4}{16} (\rho E)^4 \right] \\ \times \left(1.7 \times 10^{-3} \exp\left\{ -\frac{2\pi}{\alpha_s} \left[1 - 12 \left(\frac{4(z_0 - 2)}{z_0^3} + \frac{1}{z_0^2} \right) \right] \right\} \right. \\ \left. \times \frac{\pi^2 z_0^6}{9^{2.2} \sqrt{30} \sqrt{(z_0 + 2)(z_0 - 2.4)}} \right) \frac{d\rho}{\rho} \\ = \sigma_0 \int BL \frac{d\rho}{\rho}. \quad (5)$$

The cross section given by eq. (5) grows with ρ not only due to the factor $(\rho E)^4$ written in the explicit form, but also due to the decrease of the parameter $z_0(E\rho = (4\pi/\alpha_s) 24\sqrt{z_0 - 2/z_0^3})$ and to the increase of the running coupling constant

$$\frac{4\pi}{\alpha_s(\rho)} = b \ln \frac{1}{\rho^2 A^2}. \quad (6)$$

If one forgets about the large-size instantons ($\rho > \rho_0$) and puts $b = \frac{25}{3}$ and $A = 200$ MeV, then the cross section (eq. (5)) reaches its unitarity limit already due to the contribution of instantons with $\rho \leq \rho_0 \approx 1/(12 \text{ GeV})$ for $E = 200 \text{ GeV}$ ^{#2}. Instantons with small ρ absorb the energy of initial gluons in the S-wave and transform the main part of it into the transverse energy of secondary particles, because of the condition $E_T = \frac{1}{4}\pi E$ for an isotropic decay.

So the cross section of the events with large transverse energy may be written in the form

$$\frac{d\sigma}{dE_T^2 d\eta} = \int G(x_1, E^2) G(x_2, E^2) \delta\left(x_1 x_2 s - \frac{4^2}{\pi^2} E_T^2\right) \\ \times \delta\left(\eta - \frac{1}{2} \ln \frac{x_1}{x_2}\right) \sigma_1\left(\frac{4}{\pi} E_T\right) dx_1 dx_2, \quad (7)$$

^{#2} Of course, the integral over ρ is IR divergent. However, we want to deal with the contribution of small-size instantons which is under our control. Note that ρ_0 is not a cutoff introduced by hand but rather it reflects the region that saturates the unitary conditions. In fact, ρ_0 inversely scales with energy (cf. ref. [7]). Variation from $E = 100 \text{ GeV}$ up to $E = 500 \text{ GeV}$ leads to the variation of the product $E\rho_0$ from 13 up to $E\rho_0 = 20$ correspondingly.

where η is the rapidity of the instanton-induced cluster.

It should be pointed out that the large-size instantons ($\rho > \rho_0$) cannot change the value of the cross section $d\sigma/dE_T$ (eq. (7)). Indeed, the shadowing of the initial gluon beam by the instanton with $\rho > \rho_0$ takes place in the same S-wave and transforms into E_T more than 75% ($\frac{3}{4}\pi$) of the initial energy. The interference between instantons with different radii ($\rho_1 < \rho_0$, $\rho_2 > \rho_0$) is also suppressed, as the parameter $z = (R^2 + \rho_1^2 + \rho_2^2)/\rho_1\rho_2$ increases with the difference $|\rho_2 - \rho_1|$ and for $\rho_1 \ll \rho_2$ (or $\rho_2 \ll \rho_1$) the factor $\exp(S_{\text{int}}) \approx \exp(24\pi/\alpha_s z^2)$ becomes too small.

The analogous formula, which describes the production of a few hadron jets with large transverse energy E_T in the framework of perturbative QCD, differs from eq. (7) only in one point. One has to change the instanton-induced cross section σ_1 by the perturbative QCD cross section σ_P .

In the case of the wide-angle (90°) gluon-gluon scattering in the Born approximation one has

$$\frac{d\sigma_P}{dt} = \frac{\pi\alpha_s^2}{E^4} \cdot 30.375 \approx \frac{\alpha_s^2}{E^4} \cdot 1.9\sigma_0. \quad (8)$$

For $E_T > 100$ GeV the instanton-induced cross section $\sigma_1 \approx \sigma_0$ exceeds the perturbative value σ_P in a more than one order of magnitude. So one has to expect that the main σ_P source of the large E_T events ($E_T > 100$ GeV) should be the QCD instanton which produces heavy cluster (fireball) decaying isotropically into a large number ($\sim E\rho_0 \approx 10-20$) of gluon jets. The transverse momentum of such a jet is of the order of $q_{t,\text{jet}} \sim 1/\rho_0 \sim 10-20$ GeV.

Unfortunately, the events of this type were not observed experimentally neither at SpPb nor at FNAL colliders [8]. The values of the cross section as well as the structure (multiplicity- and $q_{t,\text{jet}}$ -distributions) of large- E_T events may be well-described in the framework of perturbative QCD [8].

Of course, this fact does not mean that the QCD instanton does not exist in nature. More probably this means that our first hypothesis, $A(\sigma_1 \rightarrow \sigma_0)$, is wrong. But at the same time one has to doubt whether it will be possible to observe baryon number violation at SSC experiments. The cross section of the instanton-induced processes is too small.

(B) Now let us consider the other extreme scenario. In accordance with the arguments of Zakharov

[4], the cross section of the instanton-induced processes increases at small energy ($E\rho \leq 1/\alpha_s$) region but decreases at larger energies $E\rho > 4\pi/\alpha_s$ ^{#3}.

Unfortunately, we do not know the realistic concrete model of such a type. So, to estimate the scale of the cross section we use the oversimplified formula that sums up the contributions of diagrams shown in fig. 1 under the conditions that both the stationary (saddle point) values of all instanton radii $\rho_i = \rho$ and all the distances between the neighbouring instantons $R_i = R$ are equal to each other.

In this case the cross section takes the Breit-Wigner form

$$\sigma_B = \sigma_0 \text{Im} \int \frac{B}{1-iL} \frac{d\rho}{\rho} = \sigma_0 \int \frac{BL}{1+L^2} \frac{d\rho}{\rho}, \quad (9)$$

where the notations are that L is used for the expression in the last large parentheses of eq. (5) and B for the expression in the first square brackets of eq. (5). Following Zakharov, one can use the analogy with the Breit-Wigner formula and say the value of L plays the role of the total width (Γ_t) and B plays the role of the elastic width (Γ_{el}) of the "resonance".

The fraction of the two gluon decay mode ($B/L \approx \Gamma_{\text{el}}/\Gamma_t$) decreases with energy and for $L > 1$ one gets

$$\sigma_B < \sigma_0 \frac{\Gamma_{\text{el}}}{\Gamma_t}.$$

Thus, the cross section conserves at least one exponentially small factor $\exp(-2\pi/\alpha_s)$ (which is contained in the B -term). In other words at $L \gg 1$ the overlap integral of the instanton (our "resonance") wave function and two initial gluons turns out to be very small, i.e. the probability to transfer the large energy of two colliding point-like gluons (partons) to the large-size instanton is suppressed by the small factor $B/L < \exp(-2\pi/\alpha_s)$.

The cross section given by eq. (9) is presented in fig. 2. There are no problems now with the divergence of the integral in the region of large ρ . The value

^{#3} The arguments in favour of this position are also given in refs. [9-11].



Fig. 1. Multi-instanton chain summed up by eq. (9).

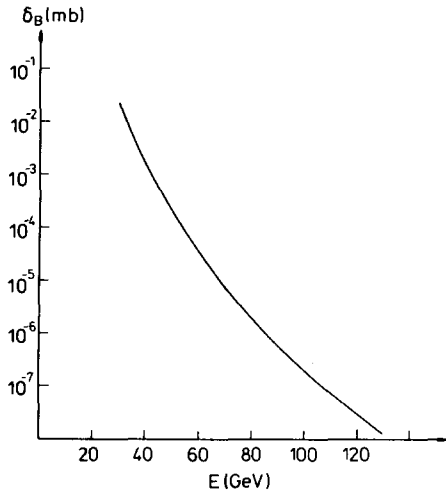


Fig. 2. Energy dependence of the instanton-induced cross section σ_B given by eq. (9). $A=200$ MeV; $b=8\frac{1}{2}$, $n_F=3$.

of L increases with ρ and the integral eq. (9) converges rapidly. The main contribution comes from the region of $z \approx 6$, where $L=1.5-1.6$. Let us note that in this range the value of L depends weakly on A_{QCD} (i.e. the coupling α_s) as at $z=6-6.1$ the coefficient "a" in front of the factor $2\pi/\alpha_s$ in the power of exponent is extremely small ($|a| < 0.02$).

Of course, in such a scenario the absolute value of the QCD instanton-induced cross section falls quickly with energy, but in the range of $E \sim 30$ GeV the instanton (σ_B) and perturbative (σ_F) QCD cross sections are of the same order of magnitude.

3. Let us fix the size of the instanton and discuss the energy dependence of the cross section σ_B eq. (9). In the region of the maximum the ratio (branching) $\Gamma_{el}/\Gamma_i = B/L$ is not too small. So one can consider a special case of quasielastic (small multiplicity) events where in addition to the six quark jets caused by the zero fermion modes the QCD instanton emits only two (or 3-4) gluon jets. At $E \sim 30$ GeV the corresponding cross section,

$$\sigma_2 = \sigma_0 \left(\int B \frac{d\rho}{\rho} \right)^2 \left\langle \left(\frac{2^4}{z^3} \right)^{n_F} \right\rangle, \quad (10)$$

contains a noticeable part of the total cross section ($\sim 0.1\%$). However, there is an important advantage. Due to small multiplicity and rather large trans-

verse momentum of jets ($\frac{1}{2}E > q_{t,jet} > 1/\rho$) the expected corrections are not too large. The confinement or the instantons with large slightly distort the distribution of large q_t jets. The influence of surrounding onto small ρ -instantons is determined by the parameter [12]

$$I = \frac{\pi^4 \rho^4}{8\alpha_s^2(\rho)} \langle 0 | \frac{\alpha_s}{\pi} G_{\mu\nu}^2 | 0 \rangle, \quad (11)$$

which does not exceed 0.1 for $\rho < 1/(2 \text{ GeV})$ (at $A=0.2 \text{ GeV}$, $\langle 0 | \alpha_s/\pi G_{\mu\nu}^2 | 0 \rangle = 0.012 \text{ GeV}^4$).

The quantum corrections to the gluon-gluon collisions in the one-instanton sector as was calculated in ref. [13] get a multiplicative factor

$$1 - \frac{1}{16\pi} \alpha_s \rho^2 E^2 \sum_{i,j} a_{ij} \ln(-a_{ij}),$$

where $a_{ij} = p_i p_j / E^2$.

Owing to the condition $\sum_i p_i = 0$ the typical values of $\text{Re} \sum_{i,j} a_{ij} \ln(-a_{ij}) \sim \frac{1}{2}$. So the quantum correction $\alpha_s/32\pi \rho^2 E^2 \leq 1$ for $\alpha_s < \frac{1}{2}$ and $\rho E < 17$.

For example, calculate the cross section of three gluon-jets (plus six quark-jets) production in gluon-gluon collision induced by the QCD instanton. To select the emission of only three gluon jets one has to expand the exponent $\exp(S_{int})$ and takes the term of the order of $(2\pi/\alpha_s)^3$. Then the cross section σ_3 will be equal to

$$\begin{aligned} \sigma_3 = & \frac{\sigma_0}{16\pi} \left[1.7 \times 10^{-3} \left(\frac{2\pi}{\alpha_s} \right)^7 \frac{\pi^2}{4} (\rho E)^2 \exp\left(-\frac{2\pi}{\alpha_s}\right) \right]^2 \\ & \times \frac{2^9}{36 \cdot 3!} \int \frac{d^4 R}{\rho^4} \exp(iER) \int \frac{d\lambda d\rho}{\rho^2 z^{15}} \\ & \times \int dU (2 \text{Tr } 0 \text{Tr } 0^+ - \text{Tr } 00^+)^5 (\text{Tr } 0 \text{Tr } 0^+)^3, \end{aligned} \quad (12)$$

where we have used again the values of $\rho = \frac{1}{2}(\rho_1 + \rho_2)$ and $\lambda = \rho_1 - \rho_2$ instead of instanton radii ρ_1 and ρ_2 . Here 0 is the 2×2 left upper corner of the matrix U of the relative instanton-anti-instanton orientation (and 0^+ is the same corner of U^+).

The integral over $d\lambda$ may be calculated by the stationary point method with rather good accuracy. The position of saddle point is $\lambda_0 = 0$. The averaging over the instanton orientations dU and integration over $d^4 R$ are taken in the explicit form. They give the fac-

tor 589 and an expression of the form of $(\rho E)^{13} K_{13}(\sqrt{2}\rho E)$, respectively. Thanks to the exponential decreasing of the McDonald function $K_{13}(\sqrt{2}\rho E) \sim \exp(-\sqrt{2}\rho E)/\sqrt{\rho E}$, the integral over ρ converges. One can interpret this convergency by the fact that in eq. (12) the amplitude of the gluon emission by the instanton is written through the parameter $1/z = \rho^2/(R^2 + 2\rho^2)$. At very large distances ($R \rightarrow \infty$) the amplitude falls as $1/R^2$ (see fig. 3a), while the corrections of the type of $(\rho^2/R^2)^n$ take into account the contributions of diagrams fig. 3b, with the same power of the coupling α_s as the graph of fig. 3a has. The interference of the one-gluon and multi-gluons cuts of the diagrams fig. 3 (see figs. 4a and 4b) which accounts for the first quantum corrections leads to the "form factor" decreasing cross section [14] $\sigma \sim \exp(-\sqrt{2}\rho E)$.

Thus, the gluon jet q_1 defined here includes all the hadrons produced in the decay of a separate gluon q_1 , or in the decay of the two neighbouring gluons q'_1 and q''_1 flying close to each other with almost the same angles, and so on.

The last integration over $d\rho$ in eq. (12) may be fulfilled by the stationary point method. Due to the growth of the running coupling $\alpha_s = 2\pi/(b \ln 1/\rho\Lambda)$ with ρ , the position of the saddle point ρ_0 , moves to the larger values of ρ , and finally one gets

$$\sigma_3 \simeq 4.9 \times 10^{-3} \sigma_0 \quad (13)$$

for $E = 30$ GeV, $\Lambda = 200$ MeV and $b = 8\frac{1}{2}$.

The essential (saddle point) values of the instanton radius and the coupling are as follows: $\rho_0^{-1} \simeq 1.9$ GeV, $\alpha_s(\rho_0) \simeq 0.335$ and the product $\rho_0 E \simeq 16$. In this case the energy of each individual jet (gluon or quark) is about 3–5 GeV.

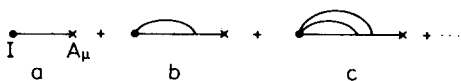


Fig. 3. Diagrams for gluon field emitted by instanton.

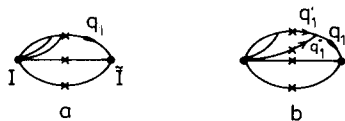


Fig. 4. Interference of different cuts of the graph of fig. 3b which contribute to the instanton-anti-instanton interaction.

For the four-gluon (+ quarks) jet production one gets at $E \simeq 30$ GeV an even larger cross section: $\sigma_4 \simeq 0.035 \sigma_0$. These instanton-induced cross sections are not too small, they exceed in more than one order of magnitude the values of the analogous cross sections (especially σ_4) calculated within perturbative QCD (the emission of each next jets gives an additional small factor $\sim N_c \alpha_s / \pi \sim \frac{1}{3}$ in the case of perturbative QCD).

Therefore, in the energy range $E \sim 30$ –60 GeV one may hope to select experimentally the small multiplicity instanton-induced events (i.e. the S-wave production of 8–9 jets by the small ρ ($\rho E \leq 16$) instanton) using such a characteristics as high sphericity, and more or less uniform distribution of energy in the secondary jets.

At lower energies E_T the transverse momenta of jets are too small. So it would be difficult to separate the QCD instanton-induced events from the ordinary soft or semihard processes. On the other hand, at larger energies the absolute values of nonperturbative cross sections σ_3 , σ_4 fall rapidly (of about three order of magnitude from $E = 30$ GeV to $E = 60$ GeV).

In spite of the fact that the corrections to the "small-multiplicity" cross sections are not very large, it is almost impossible to predict the exact value of σ_3 (or σ_4) as this value depends strongly on the QCD parameter Λ : $\sigma_3 \sim \Lambda_{\text{QCD}}^{16.67}$.

Due to this reason it is better to say that the above numerical values correspond not to the initial energy $E = 30$ GeV but to such an energy for which the saddle point coupling $\alpha_s(\rho_0) = 0.335$ (recall that for σ_3 the position of the saddle point $\rho_0 = 15.8/E$).

We would like to acknowledge the hospitality of the Department of Theoretical Physics of the University of Lund (M.G.R.) and the Department of Physics and Grey College of Durham University (I.I.B.) where part of this work was done.

References

- [1] A. Ringwald, Nucl. Phys. B 330 (1990) 1;
O. Espinosa, Nucl. Phys. B 343 (1990) 310;
L. McLerran, A. Vainshtein and M. Voloshin, Phys. Rev. D 42 (1990) 171.
- [2] A.A. Belavin, A.M. Polyakov, A.S. Schwartz and Yu.S. Tyupkin, Phys. Lett. B 59 (1975) 85.

- [3] V.V. Khoze and A. Ringwald, Phys. Lett. B 259 (1991) 106.
- [4] V.I. Zakharov, Nucl. Phys. B 353 (1991) 683.
- [5] I.I. Balitsky and A.V. Yung, Phys. Lett. B 168 (1986) 113;
A.V. Yung, Nucl. Phys. B 297 (1988) 47.
- [6] V.V. Khoze and A. Ringwald, preprint CERN-TH-6082/91.
- [7] V. Zakharov, preprint MPI-PH-92-33.
- [8] G. Arnison et al., Phys. Lett. B 123 (1983) 115;
F. Abe et al., preprint FNAL-Pub-91/283-E.
- [9] M. Maggiore and M. Shifman, Nucl. Phys. B 371 (1992) 177.
- [10] M.B. Voloshin, Nucl. Phys., B 363 (1991) 425.
- [11] N. Dorey and M.P. Mattis, Phys. Lett. B 277 (1992) 337.
- [12] M.A. Shifman, A.I. Vainshtein, V.I. Zakharov, Nucl. Phys. B 165 (1980) 45.
- [13] A.H. Mueller, Nucl. Phys. B 348 (1991) 310.
- [14] V.I. Zakharov, Nucl. Phys. B 372 (1992) 637.